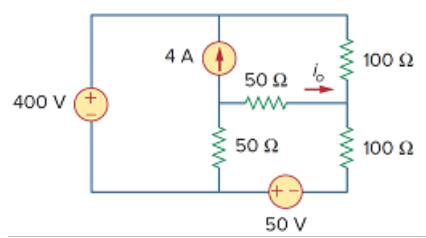


Problem

Use mesh analysis to obtain i_o in the circuit of Fig. 3.90.

Figure 3.90:



Step-by-step solution

Step 1 of 6

Refer to Figure 3.90 in the text book.

Identify the loops and currents in the circuit diagram shown in Figure 1.

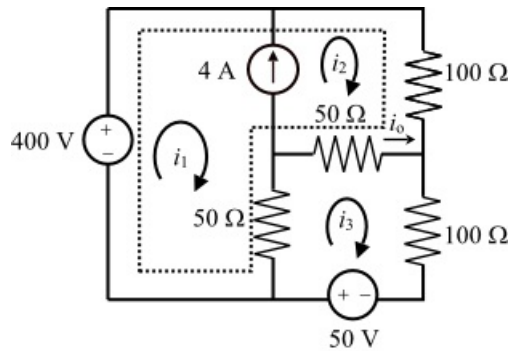


Figure 1

Step 2 of 6

Notice that loops 1 and 2 represent super mesh.

Consider the relation between the currents i_1 and i_2 .

$$i_2 - i_1 = 4$$

$$i_1 = i_2 - 4$$

Therefore, the expression for the current i_1 is, $i_2 - 4$.

Step 3 of 6

Apply super mesh analysis in loops 1 and 2.

Applying Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero.

$$-400 + i_2(100) + (i_2 - i_3)(50) + (i_1 - i_3)(50) = 0$$

$$-400 + 100i_2 + 50i_2 - 50i_3 + 50i_1 - 50i_3 = 0$$

$$-400 + 150i_2 - 100i_3 + 50i_1 = 0$$

$$-8 + 3i_2 - 2i_3 + i_1 = 0$$

Substitute $i_2 - 4$ for i_1 .

$$-8 + 3i_2 - 2i_3 + i_2 - 4 = 0$$

$$-12 + 4i_2 - 2i_3 = 0$$

$$-6 + 2i_2 - i_3 = 0$$

$$i_3 = -6 + 2i_2 \dots\dots (1)$$

Therefore, the expression for the current i_3 is $-6 + 2i_2$.

Step 4 of 6

Apply mesh analysis in loop-3.

For mesh 3, Kirchhoff's voltage law gives,

$$-50 + (i_3 - i_1)(50) + (i_3 - i_2)(50) + i_3(100) = 0$$

$$-1 + i_3 - i_1 + i_3 - i_2 + 2i_3 = 0$$

$$-1 + 4i_3 - i_1 - i_2 = 0$$

Substitute $i_2 - 4$ for i_1 and $-6 + 2i_2$ for i_3 .

$$-1 + 4(-6 + 2i_2) - (i_2 - 4) - i_2 = 0$$

$$-1 - 24 + 8i_2 - i_2 + 4 - i_2 = 0$$

$$6i_2 - 21 = 0$$

$$6i_2 = 21$$

$$i_2 = \frac{21}{6}$$

$$= 3.5 \text{ A}$$

Therefore, the value of the current i_2 is, 3.5 A .

Step 5 of 6

Recall equation (1).

$$i_3 = -6 + 2i_2$$

Substitute 3.5 A for i_2 .

$$i_3 = -6 + 2(3.5)$$

$$= -6 + 7$$

$$= 1 \text{ A}$$

Therefore, the value of the current i_3 , is 1 A .

Step 6 of 6

Calculate the value of the current i_o .

$$i_o = i_3 - i_2$$

Substitute 3.5 A for i_2 and 1 A for i_3 .

$$\begin{aligned} i_o &= 1 - 3.5 \\ &= -2.5 \text{ A} \end{aligned}$$

Therefore, the value of the current i_o is, -2.5 A.